

Solution

WORK-02

Class 09 - Science

Section A

1. The correct order of match is given as (a) – (ii), (b) – (iv), (c) – (i), (d) – (iii).
2. The correct order of match is given as (a) – (iv), (b) – (i), (c) – (ii), (d) – (iii).
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9. The correct order of match is given as (a) – (iv), (b) – (i), (c) – (ii), (d) – (iii).

Section B

10. Frequency is defined as the number of beats per second. The heart beats 75 times in 60 second, therefore its frequency is $v = \frac{75}{60} = 1.25 \text{ Hz}$

11. Sound requires medium for propagation and it can be proved by following experiment:-

- 1) Take a bell jar and suspend an electric bell in it,
- 2) The bell jar is connected to a vacuum pump. Till the air is in the bell jar, the sound of the electric bell is louder.
- 3) Now, with the help of vacuum pump, pump out the air gradually
- 4) Now as air is pumped out, the sound of the bell gets fainter and fainter.
- 5) Now, when the bell jar is completely vacuumed no sound is heard.
- 6) This shows that air is required for propagation of sound.

12. From the equation: $v = n \times \lambda$

where v = velocity/speed

n = frequency of wave

λ = wavelength of wave

13. Given wavelength, $\lambda = 0.332 \text{ m}$

Time period of wave, 10^{-3} s

Therefore, Frequency of wave, $v = \frac{1}{T}$

$$= \frac{1}{10^{-3}}$$

$$= 1000 \text{ Hz}$$

Therefore, Velocity of wave, $v = v\lambda = 1000 \times 0.332 = 332 \text{ m/s}$

In a given medium, the velocity of sound wave remains same.

New time period of new wave, $T' = 10^{-4} \text{ s}$

Frequency of new wave, $v' = \frac{1}{T'}$

$$= \frac{1}{10^{-4}}$$

$$= 10,000 \text{ Hz}$$

Therefore, Wavelength of new wave, $\lambda' = \frac{v}{v'} = \frac{332}{10,000}$

$$= 0.0332 \text{ m}$$

14. Given : Time = 2s, velocity of sound in water = 1531 ms^{-1}

Let the depth of water = d

Therefore total distance travelled by sound before it is detected by the sonar = $2d$

Using the expression distance = Velocity $\times$ Time

We have $2d = 1531 \times 2$

Therefore $d = \frac{1531 \times 2}{2} = 1531 \text{ m}$

15. For 20 Hz sound waves the wavelength would be

$$v = n \times \lambda$$

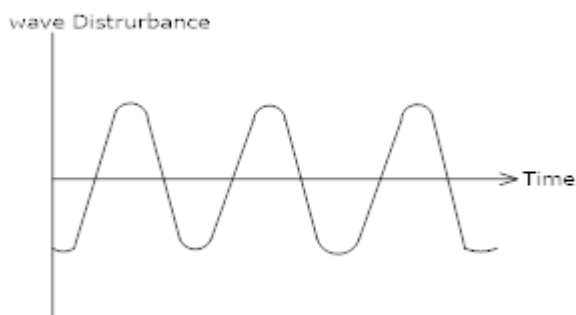
$$\lambda = v/n = 344/20 = 17.2 \text{ m}$$

For 20 kHz sound waves the wavelength would be

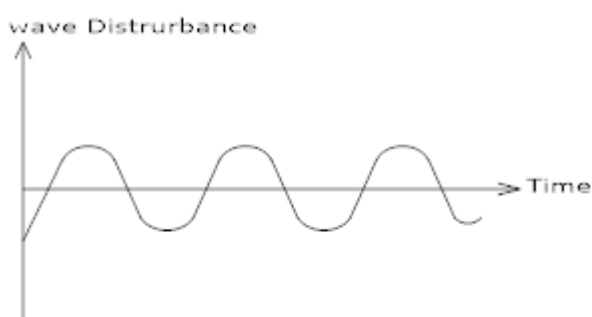
$$v = n \times \lambda$$

$$\lambda = v/n = 344 \text{ m s}^{-1} / 20000 \text{ Hz} = 0.0172 \text{ m}$$

16. As we speak, the particles of air near our mouth are pushed forward so they get compressed. Then they compress the other particles of air. As the compression proceeds, the particles of air near our mouth expand again and thus rarefaction occurs. This process is repeated further and as a result sound wave propagates in the form of compressions and rarefactions in the medium till the sound reaches your ear.
17. Louder sound: - Sound which has a higher amplitude and high energy are called louder sound.



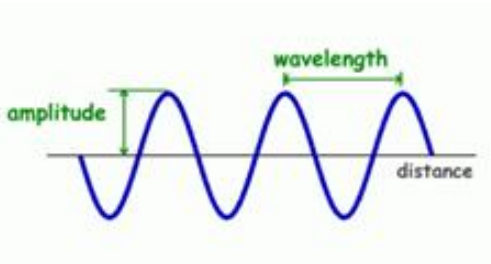
Softer sound: - Sound which has lesser amplitude and less energy are called soft sound.



18. Curved walls and ceiling of a concert hall work like concave reflecting surfaces. We know that a concave surface converges the reflected waves. This helps in ensuring that sound clearly reaches the audience.
19. wavelength: For a sound wave, the combined length of compression and an adjacent rarefaction is called its wavelength even the distance between centres of two consecutive compressions or two consecutive rarefactions is also equal to its wavelength.

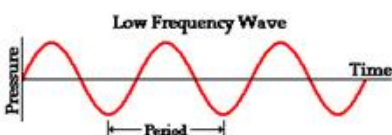
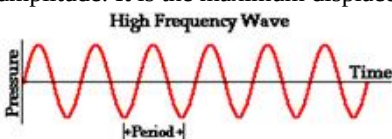


frequency: The number of vibrations or oscillations per second is called frequency i.e. it is the number of complete waves or cycles produced in one second.



time period: The time taken to complete one vibration/oscillation/complete wave is called time period. It is measured in seconds.

amplitude: It is the maximum displacement of the particles of the medium from their mean/original position at rest.



20. Given: Time of echo = 4s, velocity of sound in air = 340 ms^{-1}

Distance of cliff from child (S) = ?

Using the expression $S = \frac{V \times t}{2}$

We have $S = \frac{340 \times 4}{2} = 680 \text{ m}$

21. For the downward journey of stone.

Initial velocity (u) = 0

Distance i.e. (height) of tower (s) = 500 m

Time of fall (t_1) = ?

Acceleration due to gravity (g) = 10 ms^{-2}

We know:

$$S = ut + \frac{1}{2}at^2$$

$$500 = 0 \times t_1 + \frac{1}{2} \times 10 \times t_1^2$$

$$\Rightarrow 500 = 5t_1^2$$

$$\Rightarrow t_1^2 = 100$$

$$t_1 = 10 \text{ s}$$

For the sound travelling upward

$$\text{Time taken } (t_2) = \frac{\text{Total distance i.e. height}}{\text{Speed of sound}} = \frac{500}{340} = 1.47 \text{ s}$$

$$\text{Time required to hear splash} = t_1 + t_2 = 10 + 1.47 = 11.47 \text{ s}$$

22. As the sensation of sound persists in our brain for about 0.1 s. To hear a distinct echo the time interval between the original sound and the reflected one must be at least 0.1s. Therefore the total distance covered by the sound from the point of generation to the reflecting surface and back should be at least $(344 \text{ m/s}) \times 0.1 \text{ s} = 34.4 \text{ m}$. Thus, for hearing distinct echoes, the minimum distance of the obstacle from the source of sound must be half of this distance, that is, 17.2 m. Speed of sound will increase with increase in temperature. Therefore, on a hotter day speed of sound will be greater hence echoes may be heard more than once because of multiple reflections of sound.

23. Three features of transverse waves are:

- The particles of the medium vibrate at right angles to the direction of propagation of the wave.
- Transverse waves travel in the form of crests and troughs.
- They cannot travel through a vacuum.

24. Distance of object from submarine = 3625 m

therefore, distance travelled by sonar waves = $2 \times 3625 = 7250 \text{ m}$

since, speed = distance/time = $7250/5 = 1450 \text{ m/s}$

25. Let s be the distance of the hill from the boy and t the time of to and fro journey of sound waves, then from relation

Distance = Velocity $\times$ Time

We have,

$$2s = vt$$

$$s = \frac{vt}{2}$$

Here, $v = 340 \text{ m/s}$, $t = 0.8 \text{ s}$

$$s = \frac{340 \times 0.8}{2}$$

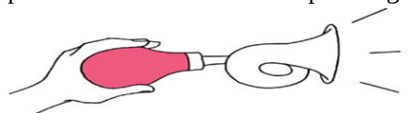
$$340 \times 0.4 \text{ m}$$

$$= 136 \text{ m}$$

26. No, I will not be able to hear sound, because Moon has no atmosphere. Therefore, no sound waves can travel to your ears and therefore, no sound is heard.

27. Two practical applications of reflection of sound waves

- Megaphones or loudhailers, horns, musical instruments such as trumpets and *shehnais*, are all designed to send sound in a particular direction without spreading it in all directions.



- A stethoscope is a medical instrument used for listening to sounds produced within the body, chiefly in the heart or lungs. In stethoscopes, the sound of the patient's heartbeat reaches the doctor's ears by multiple reflections of sound.

28. Incident line is making an angle of 50° with reflecting surface.

So, angle of incidence = $90^\circ - 50^\circ = 40^\circ$

Angle of reflection = angle of incidence

Hence, $\angle x = 40^\circ$

29. Frequency = 512 Hz, speed of the sound = 340 m/s

Since $v = \lambda u$

$\lambda = v/u = 340/512 = 0.66\text{m}$

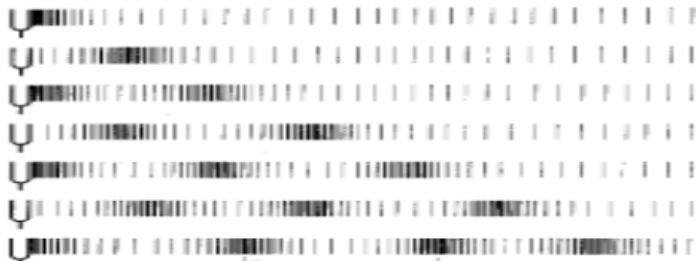
30. If the time gap between the original sound and reflected sound received by the listener is around 0.1s, only then the echo can be heard.

Velocity of sound in air = 344 m/s.

Time interval = 0.1 s

Therefore, The minimum distance travelled by the reflected sound wave for the distinctly listening to the echo = Velocity of the sound $\times$ Time interval = $344 \times 0.1 = 34.5\text{ m}$

Since, the girl is sitting in middle of a park of dimension $12\text{ m} \times 12\text{ m}$, Therefore; The distance travelled by the sound reflected from the building and then reaching to the girl = $(6 + 6) = 12\text{ m}$, which is much smaller than the required distance. Therefore, Girl will not hear any echo of sound.



31.

Sound waves are propagated because of compressions and rarefactions in the particles of the medium.

The region, where particles come closer, is called compression and region where particles move farther from each other is called rarefaction.

32. The bats produce high pitched ultrasonic waves which are not heard by human beings. The ultrasonic waves on striking the insect send back an echo, which is heard by the bat. As the echo is heard by the bat, it hovers on the insect and catches it.

33. Given: Depth of the well (d) = 44.1 m

Initial velocity of stone (u) = 0

Acceleration due to gravity (g) = 9.8 ms^{-2}

Time taken by stone to reach water level (t) = ?

Total time taken = 3.13 s

Using $S = ut + \frac{1}{2}at^2$, we have

$$44.1 = 0 \times t + \frac{1}{2} \times 9.8 \times t^2$$

$$\text{or } t^2 = \frac{44.1}{4.9} = 9$$

$$\Rightarrow t = 3.5$$

Since total time taken is 3.13 s, therefore sound takes time $t_s = 3.13 - 3 = 0.13\text{ s}$ to reach the top of the well.

$$\text{Using } V = \frac{S}{t} = \frac{44.1}{0.13} = 339.2\text{ ms}^{-1}$$

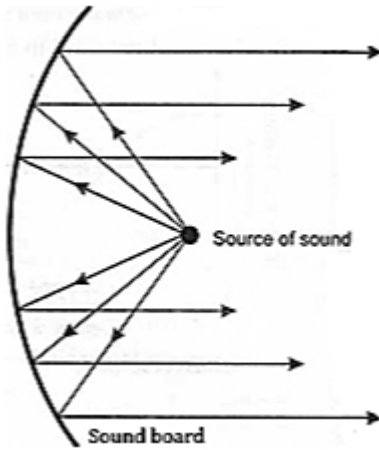
34. Mechanical waves are an oscillation of matter that requires a medium for their propagation via the mechanism of particle-to-particle interaction. Since, sound also requires a medium for its Propagation, hence it is also a mechanical wave.

35. Loudness is a measure of the response of the ear to the sound. Even when two sounds are of equal intensity, we may hear one as louder than the other simply because our ear detects it better.

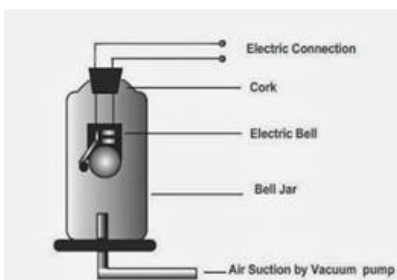
Loudness of sound depends upon the amplitude of those sound waves. Higher is the amplitude of vibrating air particles louder will be the sound.

36. The reflection of sound may take place at curved surfaces also. This fact is made use of in the large halls to spread sound evenly throughout the hall. This is done by using soundboards The speaker is located at the focus of the soundboard (Fig.) and the concave reflecting soundboards are placed behind the speakers in a large hall. The sound board prevents the spreading out of the sound waves in various directions. It sends the sound waves from the speaker at its focus, by reflection towards the audience. This

helps in making the speech readily audible even at a distance.



37. Ceilings of concert halls are corned so that sound after reflection from the curved surface which have greater surface area, can reach to all places of halls.
38. Yes. Sound follows the same laws of reflection as light does. We can say that because here the directions in which the sound is incident and is reflected make equal angles with the normal to the reflecting surface at the point of incidence, and the three are in the same plane.
39. We see the lightning before we hear the thunder because light travels faster than sound. The light from the lightning travels to our eyes much quicker than the sound from the lightning. so we hear it later than we see it.
40. Take an electric bell and an airtight glass bell jar. The electric bell is suspended inside the airtight bell jar. The bell jar is connected to a vacuum pump. If you press the switch you will be able to hear the bell. Now start the vacuum pump. When the air in the jar is pumped out gradually, the sound becomes fainter, although the same current is passing through the bell. After some time when less air is left inside the bell jar, you will hear a very feeble sound. Now if we evacuate the bell jar no sound is heard.



Bell jar experiment showing sound needs a material medium for its propagation.

Result: The above-mentioned activity shows that sound needs a medium to propagate.

41. Let the time period of a wave be T seconds.
 In T seconds, number of waves generated = 1.
 So, in 1 second number of waves generated = $\frac{1}{T}$
 But number of waves generated in 1 second is frequency.
 Therefore, $v = \frac{1}{T}$
 Now, Velocity = $\frac{\text{Distance travelled}}{\text{Time taken}}$
 $= V = \frac{\lambda}{T}$
 $= V = \frac{1}{T} \cdot \lambda$
 $V = v\lambda$
42. a. The amplitude of the wave determines the loudness. The more the amplitude of a wave, more is the loudness produced.
 b. The pitch is determined by the frequency of the wave. The higher the frequency of a wave, more is its pitch and shriller is the sound.
43. Given: $V_a = 340 \text{ ms}^{-1}$ and $V_{\text{iron}} = 5130 \text{ ms}^{-1}$
 We know that $V = \frac{\text{distance travelled}}{\text{time}}$
 $\Rightarrow t = \frac{\text{distance travelled}}{\text{velocity}}$

Let length of the pipe = L

Now time taken by sound to travel through air

$$t_a = \frac{L}{340} \dots (1)$$

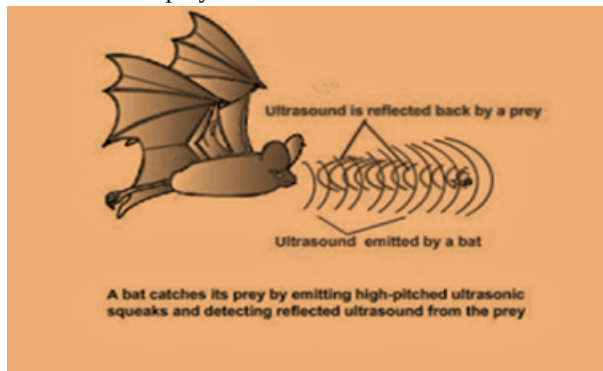
Also, time taken by the sound to travel through iron

$$t_{\text{iron}} = \frac{L}{5130} \dots (2)$$

Therefore the ratio of time taken in air to that taken in iron is obtained by dividing equation 1 by equation 2, thus

$$\frac{t_a}{t_{\text{iron}}} = \frac{L}{340} \times \frac{5130}{L} = 15.08 \text{ which is approximately 15 times.}$$

44. When the peon strikes the school bell with a hammer, the particles of bell metal start vibrating and those vibrations produce sound.
45. Bats search out prey and fly in dark night by emitting and detecting reflections of ultrasonic waves. The high-pitched ultrasonic squeaks of the bat are reflected from the obstacles or prey and returned to bat's ear. The nature of reflections tells the bat where the obstacle or prey is and what it is like.



46. The speed of light is $3 \times 10^8 \text{ ms}^{-1}$ whereas that of sound is 344 ms^{-1} in air. Thus, flash of lightning is seen at once, but sound takes few seconds to reach the ears.
47. The relationship between the velocity of the sound, its wavelength, and frequency is given by the formula $v = \lambda \times T$
Using the formula

Given velocity, $v = 340 \text{ ms}^{-1}$

i. $340 = 256 \lambda$

frequency, $\lambda = \frac{340}{256}$
 $= 1.33 \text{ m}$

ii. $340 = v (0.85)$

wavelength, $v = \frac{340}{0.85}$
 $= 400 \text{ Hz.}$

48. The frequency of the vibrating pendulum does not lie within the audible range (20 Hz to 20,000 Hz) and hence it does not produce audible sound.
49. Case (i) : When the frequency of sound is 20 Hz
Using the expression $V = v \times \lambda$ we have
 $\lambda = \frac{V}{v} = \frac{340}{20} = 17 \text{ m}$
 Case (ii) : When the frequency is 20,000 Hz
Using the expression $V = v \times \lambda$ we have
 $\lambda = \frac{V}{v} = \frac{340}{20000} = 0.017 \text{ m} = 1.7 \text{ cm}$
50. If the temperature rises the speed of sound will increase. This in turn will increase the minimum distance required for hearing an echo. No echo is heard because, the distance between the source of sound and reflecting body does not increase.
51. We have given that,

The time between transmission and detection, $t = 3.42 \text{ s}$

Speed of ultrasound in seawater, $v = 1531 \text{ m/s}$

Therefore, Distance travelled by the ultrasound

$$= 2 \times \text{depth of the sea} = 2d$$

where d is the depth of the sea.

$$2d = \text{speed of sound} \times \text{time}$$

$$= 1531 \text{ m/s} \times 3.42 \text{ s} = 5236 \text{ m}$$

$$d = \frac{5236}{2} \text{ m} = 2618 \text{ m}$$

Thus, the distance of the seabed from the ship is 2618 m or 2.62 km.

52. The relation between wave velocity (v), frequency (?) and wavelength (λ) is

$$v = v\lambda$$

Given, $\lambda = 1 \text{ cm}$

$$= 1 \times 10^{-2} \text{ m}, v = 330 \text{ m/s}$$

Therefore, Frequency, $\nu = \frac{v}{\lambda}$

$$\frac{330}{1 \times 10^{-2}}$$

$$= 33,000 \text{ Hz.}$$

This frequency is more than 20,000 Hz (maximum frequency audible to man). Hence, it cannot be heard by a normal human being.

53. Given frequency, $\nu = 2 \text{ kHz} = 2 \times 10^3 \text{ Hz}$,

Wavelength, $\lambda = 40 \text{ cm} = 0.40 \text{ m}$

Speed of sound = Frequency $\times$ Wavelength

i. e., $v = v\lambda$

$$= (2 \times 10^3 \text{ Hz}) \times (0.40 \text{ m})$$

$$= 0.80 \times 10^3 \text{ Hz} = 800 \text{ ms}^{-1}$$

Time, $t = \frac{\text{Distance}}{\text{Speed}}$

$$t = \frac{s}{v}$$

Given distance, $s = 1.6 \text{ km} = 1.6 \times 10^3 \text{ m}$

$$\text{Time, } t = \frac{1.6 \times 10^3 \text{ m}}{800 \text{ ms}^{-1}}$$

$$= \frac{1600}{800} \text{ s} = 2 \text{ s}$$

54. We have given that,

Frequency, $\nu = 2 \text{ kHz} = 2000 \text{ Hz}$

Wavelength, $\lambda = 35 \text{ cm} = 0.35 \text{ m}$

We know that speed, v of the wave = wavelength $\times$ frequency

$$v = \lambda \nu$$

$$= 0.35 \text{ m} \times 2000 \text{ Hz} = 700 \text{ m/s}$$

The time taken by the wave to travel a distance, d of 1.5 km is

$$t = \frac{d}{v} = \frac{1.5 \times 1000 \text{ m}}{700 \text{ ms}^{-1}} = \frac{15}{7} \text{ s} = 2.1 \text{ s}$$

Thus sound will take 2.1 s to travel a distance of 1.5 km.

55. The time interval between successive compressions from the source

$$T = 1/\nu = 1/500 = 0.002 \text{ second.}$$

56. From the graph, It is clear that

$$\text{Time period, } T = 2 \times 10^{-6} \text{ s.}$$

$$\text{Frequency, } \nu = \frac{1}{T} = \frac{1}{2 \times 10^{-6}} = 5 \times 10^5 \text{ Hz.}$$

Velocity = 1500 m/s

$$\text{Wavelength, } \lambda = \frac{5 \times 10^5}{1500} = 333.3 \text{ m.}$$

57. Distance travelled by a sonar pulse = speed of sound in salt water $\times$ time

$$= 1531 \times 1.02 = 1561.62 \text{ m}$$

$$\text{therefore, the distance of cliff from submarine} = 1561.62/2 = 780.81 \text{ m}$$

58. The speed of sound wave = 339 ms^{-1}

Wavelength of sound wave = 1.5 cm = 0.015 m

$$\therefore \text{Frequency of sound wave} = \frac{\text{Speed of sound}}{\text{Wavelength}} = \frac{339}{0.015} = 22600 \text{ Hz}$$

The sound will not be audible, because a human being can hear only up to 20,000 Hz.

59. Frequency of source of sound being 100 Hz means the sound source vibrates 100 times in one second.

$$\text{therefore vibrations made by sound source in 1 min (60 sec)} = 100 \times 60 = 6000$$

60. SONAR = Sound Navigation And Ranging.

A transmitter producing ultrasonic waves is fitted at the bottom of a ship or a boat. The ultrasound waves emitted by the transmitter go to the bottom of the sea and get reflected from the bottom. These are received back by a detector also fitted at the bottom. Knowing the time elapsing between sending and receiving back of the ultrasonic waves and the speed of these waves in water, the depth of sea can be calculated.

61. When sound waves strike a surface, they are reflected with the same velocity and the reflected sound waves are heard as echo. It is used in SONAR and detecting flaws in metal objects.
62. The sound produced by the humming bees is within the audible range of humans, i.e. between 20 Hz to 20,000 Hz. But the sound produced by vibrations of pendulum is not within our hearing range. Due to this, we are able to hear the sound of a humming bee but unable to hear the sound of a pendulum.
63. a) Time Period of a wave(T) :- The time required by a wave to complete one vibration is called time Period.
b) Amplitude(A) :- The magnitude of the maximum disturbance in the medium on either side of mean value is called the amplitude of wave.
64. Reverberation is the repeated multiple reflections of sound in any big enclosed space. It can be reduced by covering the ceiling and walls of the enclosed space with some absorbing materials like fibre board, loose woollens etc.
65. Given: $v = 20 \text{ Hz}$, $V = 30 \text{ cms}^{-1}$
Now minimum distance of separation between two consecutive compressions = Wave length = ?
Using $V = v \times \lambda$, we have $\lambda = \frac{V}{v} = \frac{30}{20} = 1.5 \text{ cm}$
66. Ultrasound is generally used to clean parts located in hard-to-reach places, for example, spiral tube, odd shaped parts, electronic components etc. Objects to be cleaned are placed in a cleaning solution and ultrasonic waves are sent into the solution. Due to the high frequency, the particles of dust, grease and dirt get detached and drop out. The objects thus get thoroughly cleaned.
67.
 - Sound intensity is a property of the sound source but loudness depends on the sound source, the medium and the receiver, as well.
 - Sound intensity holds a small significance in problems involving human hearing system, but loudness is a very important property to consider in such problems.
 - Sound intensity is measured in Watt per square meter whereas loudness is measured in Sones.
68. Given: Distance between two consecutive crests = Wave length = 100 m
Wave velocity = 20 ms^{-1}
Using the expression $V = v \times \lambda$ we have
 $v = \frac{V}{\lambda} = \frac{20}{100} = 0.2 \text{ Hz}$
69.
$$\frac{\text{Time taken by sound travel aluminium}}{\text{Time taken by sound in air}} = \frac{\text{Speed of sound in air}}{\text{Speed of sound in aluminium}}$$

Speed of sound in air Speed of sound in aluminium
$$\Rightarrow \frac{T_{\text{alu}}}{T_{\text{air}}} = \frac{346}{6420}$$

$$\Rightarrow T_{\text{Alu}} : T_{\text{Air}} = 346:6420 = 1: 18.55$$
70. The ultrasound waves are allowed to pass through metal block to which detectors are fitted. If there is a small defect in the metal block like an air bubble or a crack, then the ultrasound waves are reflected from such spots. Metal block if defective is indicated by the reflected ultrasonic waves.
71. Bats, have special types of wings. When they fly they produce ultrasonic waves. These waves are received by the ears of bat after they have been reflected by the object. The ears of the bat are so sensitive and trained that they not only get information of distance of the obstacle but also that of the nature of the reflecting surface.
72. Given, wavelength, $\lambda = 20 \text{ cm} = 0.20 \text{ m}$, wave velocity, $v = 24 \text{ ms}^{-1}$
(a) From the relation, $v = v\lambda$
$$v = \frac{v}{\lambda}$$

$$\frac{24}{0.20} \text{ waves per second}$$

(b) Time period, $T = \frac{1}{v} = \frac{1}{120} \text{ second}$
$$= 8.33 \times 10^{-3} \text{ seconds.}$$